33 Mechanisms in Heterogeneous Catalysis

Why Are We Learning This?

In previous lectures, we introduced catalytic reactions and learned that heterogeneous catalysis occurs through a sequence of adsorption, surface reaction, and desorption steps. However, a catalyst mechanism often contains many elementary reactions, and not all of those reactions influence the overall rate equally. The challenge is determining:

Which Elementary Steps Occur? Which Step Controls the Rate? What Mechanism Explains
the Observed Rate Law?

These questions are at the heart of catalytic reaction engineering. By identifying the reaction mechanism and rate-limiting step, engineers can optimize catalyst composition, operating conditions, and reactor performance.

Course Roadmap

Introduction to CRE
↓
General Balance Equation
↓
Common Reactor Models
(CSTR, Batch, PFR)
↓
Reaction Progress
(Conversion)
↓
Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)
↓
Rate Laws
(Kinetics)
↓
Equilibrium
(Reversible Reactions)

↓
Equilibrium Conversion
(Thermodynamic Limitations)
↓
One Reaction: Solving in terms of Reaction Progress
↓
One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)
↓
Catalytic Reactions
↓
Numerical Methods
(Python)
↓
Energy Balances
↓
One Reaction: Non-Isothermal Reactor Design
↓
Multiple Reactions
↓
Reactor Safety
↓
Challenging the Well-Mixed Assumption
(Residence Time Distributions)
↓
►Challenging the Elementary Rate Law Assumption◄
(Reaction Mechanisms)

Learning Objectives

By the end of this lecture, you should be able to:

- Define a catalytic reaction mechanism.
- Define a rate-limiting step.
- Explain why adsorption and desorption are often treated as equilibrium processes.

- Define surface coverage.
- Write site balances.
- Relate reaction mechanisms to observed rate laws.
- Use equilibrium assumptions to eliminate surface species.
- Test whether a proposed mechanism is consistent with experimental observations.

Big Idea

Experimental Rate Law
↓
Proposed Mechanism
↓
Rate-Limiting Step
↓
Predicted Rate Law

A successful mechanism must reproduce the experimentally observed kinetics.

Consider the reaction A (g) --> B (g), which is happening over a catalyst:

Steps:

1. Adsorption of A
2. Surface reaction: A --> B
3. Desorption of B

What Is a Reaction Mechanism?

A reaction mechanism is the sequence of elementary steps that converts reactants into products.

For a catalytic reaction:

Adsorption
↓
Surface Reaction
↓
Desorption

the overall reaction may appear simple, but multiple molecular events occur on the catalyst surface before products are formed.

Key Insight

Overall Reaction
↓
What We Observe

Mechanism
↓
How It Happens

Reaction "mechanism": Series of steps taken to get from reactants to products.

Mechanism in this example:

* are catalytic sites *'ed species are bound to the catalyst

Adsorption and desorption steps are often "unactivated." This means

Why Are Surface Reactions Often the Slow Step?

In many catalytic systems, adsorption and desorption occur relatively rapidly, while the actual chemical transformation on the catalyst surface requires bond breaking and bond formation and is therefore often slower.

Relative Speeds of the Steps

Adsorption: Fast
Surface Reaction: Often Slow
Desorption: Fast

Because the surface-reaction step frequently has the largest activation barrier, it is often assumed to be the rate-limiting step.

Physical Interpretation

Reactants Reach Surface
↓
Rapidly Adsorb
↓
Surface Chemistry Occurs
↓
Products Form
↓
Rapidly Desorb

Much of the catalytic cycle is spent waiting for the surface reaction to occur, while adsorption and desorption adjust relatively quickly.

Why This Matters

If the surface reaction is the slowest step, then:

Surface Reaction Rate
↓
Controls Overall Rate

This allows us to simplify the mechanism by treating adsorption and desorption as equilibrium processes while focusing our analysis on the surface-reaction step.

Key Insight

Fast Steps
↓
Near Equilibrium

Slow Step
↓
Controls Overall Rate

Identifying the slowest step is often the first step in developing a catalytic reaction mechanism and deriving a rate law.

The surface reaction is thus often the

Rate limiting step:

What Is a Rate-Limiting Step?

The rate-limiting step is the slowest elementary step in the mechanism.

Physical Interpretation

Fast Step
↓
Fast Step
↓
SLOW STEP
↓
Fast Step

The slow step controls how quickly material can move through the entire mechanism.

Analogy: Traffic Flow is controlled by:

Bottleneck

not by the fastest section of highway. Likewise, reaction rate is controlled by the bottleneck step in the mechanism.

Deriving mechanisms for catalytic reactions:

- It is useful to know the reaction mechanism and rate limiting step, because it allows you to tune the catalyst properties to carry out the reaction more efficiently

- One way to estimate the mechanism is to run an experiment to get the rate law, and then devise a mechanism and rate limiting step that explain the experimental observations

- Actual research example: 2 NO (g) + O_2 (g) --> 2 NO_2 (g) over a Pt catalyst

NOTE: This is an actual research example that demonstrates "exceptions to the rule" and complexities that are usually intentionally omitted from textbook and exam problems.

Observed rate law: $-r_{NO} = \frac{kC_{NO}C_{O_2}}{C_{NO_2}}$, i.e., 1st order in NO and O_2 and -1st order in NO_2

We want to devise a mechanism and rate limiting step that explain this behavior. What steps could we/should we include?

Why Do We Care About Mechanisms?

Knowing the mechanism allows engineers to understand why a catalyst performs the way it does. If the rate-limiting step is identified, catalyst properties can be modified to accelerate that step.

Engineering Workflow

Measure Rate Law
↓
Propose Mechanism
↓
Identify Bottleneck
↓
Improve Catalyst

Mechanisms therefore provide a path from experimental observations to catalyst design.

What Can a Rate Law Tell Us?

The observed rate law contains clues about what is happening on the catalyst surface. For example: Positive Reaction Order usually suggests a species promotes the reaction. Negative Reaction Order often suggests a species blocks catalytic sites.

Example

For NO oxidation: Positive Dependence on NO, Positive Dependence on O2, and Negative Dependence on NO2 suggests that NO2 may occupy surface sites that are needed for reaction.

Key Idea

Rate Law

↓
Provides Evidence About Surface Chemistry

Steps:

Recall: * = catalyst sites
*'ed species are bound to the catalyst

Note: "sites," i.e., combinations of *'s and *'ed species, must balance in each step.

How do we get the rate from this mechanism?

The rate of the reaction ≈ the rate of the rate limiting step.

Which step should we assume is the rate limiting step?

Let's choose step

If step is the rate limiting step, then all other steps (including step) are assumed to be in equilibrium. I.e., the net rates of all other steps are equal to 0.

Why Do We Assume Equilibrium for the Other Steps?

When one elementary step is significantly slower than all of the others, that step controls the overall reaction rate.

Fast Step
↓
Fast Step
↓
Slow Step
↓
Fast Step

Because the remaining steps occur much more rapidly, they are able to adjust almost immediately to changes caused by the slow step. As a result, these faster steps are often approximated as being in equilibrium.

Physical Interpretation

Slow Step Controls Overall Rate
↓
Fast Steps Respond Quickly
↓
Fast Steps Remain Near Equilibrium

This approximation is widely used in catalytic reaction engineering because it greatly simplifies the analysis of complex mechanisms.

Why This Helps

Equilibrium relationships allow us to express:

Surface Coverages in terms of measurable quantities such as:

Gas Pressures, Gas Concentrations

rather than quantities that are difficult to measure directly.

Key Insight

Assume Slow Step Controls the Rate
↓

Treat Other Steps as Equilibrium
↓
Solve for Surface Coverages
↓
Derive Overall Rate Law

This approach allows us to connect a proposed reaction mechanism to the experimentally observed rate law and test whether the mechanism is consistent with the data.

Under our assumption, the rate of NO consumption =

Therefore, $-r_{NO} =$

Here θ_i are "coverages," i.e., $\frac{\text{\# of catalytic species occupying a catalytic site}}{\text{\# of catalytic sites}}$.

By definition, $0 \le \theta \le 1$, and $\sum_i \theta_i = 1$.

What Is Surface Coverage?

Surface coverage describes the fraction of catalyst sites occupied by a particular species.

Interpretation: θNO = Fraction of Sites Occupied by NO

Limits:

0 means: No Sites Occupied

1 means: All Sites Occupied

Why Coverage Matters

Surface reactions occur only on occupied catalytic sites. The reaction rate therefore depends directly on surface coverage.

The Site Balance

Catalytic sites are conserved. Every site must be either Empty or Occupied by some surface species. Therefore: Sum of All Coverages = 1

Physical Interpretation

Empty Sites + NO-Covered Sites + O-Covered Sites + NO2-Covered Sites = All Sites. The site balance is one of the most important equations in catalytic reaction engineering

Let's test our assumption that step is the rate limiting step. We need:

It is difficult to measure the coverages of catalytic species under reaction conditions, so instead, we need expressions for them.

Key Challenge in Catalysis

Most catalytic intermediates exist only on the catalyst surface and are present in very small amounts. Under reaction conditions, these surface species are often difficult—or even impossible—to measure directly. Instead, we use: Rate Laws, Equilibrium Relationships, Site Balances, and Reaction Mechanisms to estimate surface coverages and reaction intermediates indirectly.

Engineering Reality

Surface Species
↓
Difficult to Measure Directly
↓

Build Mechanistic Model
↓
Infer Surface Coverages and Reaction Pathways

Because many of the most important catalytic species cannot be observed directly, catalytic reaction engineering relies heavily on mathematical modeling to connect measurable quantities (such as reaction rates and gas compositions) to the molecular processes occurring on the catalyst surface.

Key Insight

What We Measure
↓
Gas Composition and Reaction Rate

What We Want to Know
↓
Surface Coverages and Elementary Steps

Much of catalysis research involves using experimental observations to infer what is happening on the catalyst surface.

Summary and Key Takeaways

Today we examined how reaction mechanisms are developed for heterogeneous catalytic systems.

Key Ideas

☑ A catalytic mechanism consists of multiple elementary steps.

☑ Adsorption and desorption are often much faster than surface reactions.

☑ The rate-limiting step acts as the bottleneck for the overall mechanism.

☑ Experimental rate laws provide clues about the underlying mechanism.

☑ Surface coverage describes the fraction of catalyst sites occupied by a species.

☑ Site balances are essential for relating coverages to measurable quantities.

☑ Fast steps are often approximated as being in equilibrium.

☑ A valid mechanism must reproduce the experimentally observed rate law.

Most Important Idea

Observed Rate Law
↓
Provides Clues
↓
Reaction Mechanism
↓
Rate-Limiting Step
↓
Catalyst Design

The ultimate goal of mechanism development is to connect experimental observations with molecular-level surface chemistry so that catalysts can be designed and improved rationally.